

for gain,

$$A_v = (g_{m1} r_{o1}) (g_{m2} r_{o2})$$

$$A_{v0} = \frac{1}{\lambda_n V_{eff,12} \lambda_p V_{eff,8}}$$

$$= \frac{1}{160 \times 10^{-3} \times 0.2 \times 190 \times 10^{-3} V_{eff}} > 10^{40/20} \text{ dB}$$

$$0.520 > V_{eff,8}$$

$$\text{Let } V_{eff,8} = 0.2 \text{ V}$$

Assuming that all transistors have a

$$V_{DS(sat)} = V_{OV} = V_{eff} = 0.2 \text{ V}$$

Finding the widths and lengths of all the transistors.

$$\text{Let } (\mu_n C_{ox})_p = 47.2 \text{ } \mu\text{F/V}^2$$

$$(\mu_n C_{ox})_n = 167 \text{ } \mu\text{F/V}^2$$

for standard $0.5/0.18$ dimensions.

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{GS} - V_t)^2$$

for M_1, M_2

$$I_D = 18.849 \text{ } \mu\text{A}$$

for transistors,

$$18.849 = \frac{167}{2} \frac{W}{L} (0.2)^2$$

$$5.6434 = \frac{W}{L}$$

$$2 \times 0.5/0.18 = \frac{W}{L}$$

$$\boxed{W_{1,2} = 1 \mu\text{m}}$$

for biasing circuit transistors,

M_5, M_6, M_7

$$I_{BIAS} = I_{tail} = 2 I_D = 37.699 \times 10^{-6} \text{ A}$$

nmos - bias circuit

$$37.699 \text{ } \mu\text{A} = \frac{167}{2} \frac{W}{L} (0.2)^2$$

$$11.287 = \frac{W}{L}$$

$$4.063 \times (0.5/0.18) = \frac{W}{L}$$

$$\boxed{W_{5,6,7} = 2 \mu\text{m}}$$

for p-mos load transistors:

$$I_D = 18.849 \text{ } \mu\text{A}$$

$$18.849 \text{ } \mu\text{A} = \frac{(\mu_n C_{ox})_p}{2} \frac{W}{L} (V_{GS} - V_t)^2$$

$$18.849 \text{ } \mu\text{A} = \frac{47.2}{2} \frac{W}{L} (0.2)^2$$

$$19.967 = \frac{W}{L}$$

$$7.188 \times \left(\frac{0.5}{0.18}\right) = \frac{W}{L}$$

$$\boxed{W = 4 \mu\text{m}}$$

RHP Zero Calculation:

$$\omega_{z,RHP} = \frac{g_{m8}}{C_c}$$

$$\omega_{z,RHP} = \frac{376.99 \times 10^{-6}}{10 \times 10^{-12}}$$

$$\omega_{z,RHP} = 37.699 \times 10^6$$

$$\boxed{\omega_{z,RHP} = 37.699 \text{ MHz}}$$

Calculating g_{m8} ,

$$g_{m8} = \frac{2 I_D}{V_{ov}}$$

$$g_{m8} = \frac{2 \times 37.699}{0.2}$$

$$\boxed{g_{m8} = 376.99 \text{ } \mu\text{S}}$$

Designing the Compensation for the RHP zero,

$$R_z = \frac{1}{g_{m8}}$$

$$= \frac{1}{g_{m8}}$$

$$\boxed{R_z = 2652.59 \text{ } \Omega}$$

Let $R_z = 3 \text{ k}\Omega$

Equivalent resistance = $3 \text{ k}\Omega$

$$\frac{1}{R_z} = \frac{1}{R_n} + \frac{1}{R_p}$$

$$\frac{1}{3 \text{ k}} = \frac{2}{R_{n,p}}$$

$$R_{n,p} = 6 \text{ k}$$

Finding $\left(\frac{W}{L}\right)$ of M_9 , M_{10} .

$$R_n = 6k\Omega$$

$$6k\Omega = \frac{1}{(\mu_n C_{ox})_n \times \frac{W}{L} (0.2)^2}$$

$$\frac{W}{L} = \frac{1}{167 \mu \times 6k\Omega \times (0.2)^2}$$

$$\frac{W}{L} = 24.95$$

$$\frac{W}{L} \cong 25$$

$$\frac{W}{L} = 9 \times \frac{0.5}{0.18}$$

$$\boxed{\frac{W}{L} = 4.5 \mu.}$$

$$R_p = 6k\Omega. \text{ (PMOS dimensions).}$$

$$6k\Omega = \frac{1}{(\mu_n C_{ox})_p \times \frac{W}{L} (0.2)^2}$$

$$\frac{W}{L} = \frac{1}{47.2 \times 10^{-6} \times (0.2)^2 \times 6000}$$

$$\frac{W}{L} = 88.276$$

$$\frac{W}{L} = 31.7 \times \left(\frac{0.5}{0.18}\right)$$

$$\boxed{W_{10} = 18.75 \mu.}$$

The RHP zero cancellation resistor is implemented using a parallel NMOS-PMOS tode structure to improve linearity and reduce signal dependence. Each device is designed to contribute approximately $6k\Omega$ resulting in effective resistance of $3k\Omega$

For ~~the~~ primary pole,

$$C_{eq} = C_c (1 + g_{m2} r_{o2})$$

$$= 10pF (1 + g_{m2} r_{o2})$$

We know that

$$g_{m2} = 188.49 \mu A/V$$

$$r_{o2} = \frac{1}{\lambda_n \frac{I_D}{2}} \quad \left\{ \begin{array}{l} \lambda_n = 160 mV^{-1} \\ \lambda_p = 190 mV^{-1} \end{array} \right.$$

$$\frac{I_D}{2} = 18.849 \mu A$$

$$r_{o2} = \frac{1}{160 \times 10^{-3} \times 18.849 \mu A}$$

$$\boxed{r_{o2} = 331582.5713}$$

$$r_{o4} = \frac{1}{\lambda_p \frac{I_D}{2}}$$

$$= \frac{1}{190 \times 10^{-3} \times 18.849 \times 10^{-6}}$$

$$\boxed{r_{o4} = 279227.4335}$$

$$\# r_{os} = \frac{1}{\lambda_p D_D}$$

$$r_{os} = \frac{1}{190 \times 10^{-3} \times 2 \times 18.849 \times 10^{-6}}$$

$$r_{os} = 139613.7168$$

$$Pole_1 = \frac{1}{(331582.5113) \parallel (279227.43) \cdot 10 \times 10^{-12} \cdot (1 + [376.99 \times 10^{-6} \times 139613.7168])}$$

$$= \frac{1}{(151580.605 \times 10 \times 10^{-12}) \cdot (53.6329)}$$

$$\omega_{p1} = 12.3 \text{ kHz}$$

$$GBW = \frac{g_m}{C_c \times 2\pi}$$

$$= \frac{376.99 \times 10^{-6}}{10 \times 10^{-12} \times 2\pi}$$

$$= 6 \text{ MHz}$$

secondary pole.

$$p_2 \approx \frac{1}{(r_{o1} \parallel r_{os} \parallel R_L)(C_L + C_c)}$$

$$r_{o1} = \frac{1}{\lambda_n D_D} = \frac{1}{160 \times 10^{-3} \times 37.699 \times 10^{-6}}$$

$$= \frac{1}{6.031 \times 10^{-6}} = 165786.89$$

$$p_2 = \frac{1}{(10 \text{ k}\Omega \parallel (165786.89) \parallel (139613.716)) (20 \times 10^{-12} + 10 \times 10^{-12})}$$

$$= \frac{1}{8834.3546 \times 30 \times 10^{-12}} = 3.773 \text{ MHz}$$

$$\omega_{p1} = 12.3 \text{ KHz}$$

$$\omega_{p2} = 3.713 \text{ MHz}$$

$$\omega_z = g_{m2}/C_c =$$

$$= \frac{\cancel{376.99 \times 10^{-6}}}{\cancel{10 \times 10^{-12}}} = \frac{188.4955 \times 10^{-6}}{10 \times 10^{-12}} = 18.84 \text{ MHz}$$

$$= \cancel{376.99 \text{ MHz}}$$

$$\text{Phase Margin} = 90^\circ - \tan^{-1}(6 \text{ MHz} / 3.713 \text{ MHz}) - \tan^{-1}(6 \text{ MHz} / \cancel{376.99 \text{ MHz}})$$

$$= 90^\circ + 57.83^\circ$$

$$- 17^\circ$$

$$= +130.83^\circ$$

$$\text{PM} = 180^\circ - 130.83^\circ$$

$$= 49.17^\circ$$

Using Compensation technique of using Resistors.

$$R_z = 3 \text{ k}\Omega$$

$$\omega_z \text{ is shifted to } (1 - g_{m2} R_z) / (R_z C_c)$$

$$= (1 - (188.49 \times 10^{-6} \times 10^3 \times 3)) / (3 \times 10^3 \times 10 \times 10^{-12})$$

$$= 14.48483 \text{ MHz}$$

$$\text{Phase margin} = 90^\circ + 57.83^\circ - \cancel{17^\circ} - 22.50^\circ$$

$$= \cancel{138.33} - 125.32^\circ$$

$$= 180^\circ - \cancel{138.33} = \cancel{46.67} = \cancel{17^\circ}$$

$$= 54.67^\circ$$

With the zero shifted to 14.48 MHz from 18.84 MHz, the phase lead at unity gain increases resulting in a phase margin of 55° .

Simulation for pre-lab showing the gain-margin, phase margin ,gain and phase:

